UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. Which structure chooses between two alternative paths?
   A. array
   B. if...else
   C. struct
   D. for

2. What is displayed by the following code?

```
for (int i = 1; i <= 4; i++)
    cout << i * 2;
```

   A. 246
   B. 1357
   C. 2468
   D. 1234

3. What does the % operator return?
   A. The quotient
   B. The absolute value
   C. The square root
   D. The remainder of integer division

4. Which phase of the program development life cycle defines what the program must do?
   A. Analysis
   B. Coding
   C. Testing
   D. Maintenance

5. Which of the following declarations are valid?
   i. int a[5];
   ii. int b;
   iii. int c[3] = {1, 2, 3};
   iv. int d[];
   A. ii and iv
   B. iii and iv
   C. i and ii
   D. i and iii

6. Which input pattern stops when the user enters -1?
   A. A sentinel-controlled loop
   B. A function prototype
   C. A fixed for loop
   D. A switch statement

7. Which term describes a finite sequence of precise steps that solves a problem?
   A. Variable
   B. Algorithm
   C. Compiler
   D. Array

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8. Which data type stores a whole number?
   A. string
   B. char
   C. int
   D. float

9. Which of the following are selection structures?
   i. if
   ii. switch
   iii. for
   iv. while
   A. ii and iv
   B. i and iii
   C. iii and iv
   D. i and ii

10. Which declaration creates a floating-point variable named price?
   A. float price;
   B. int price;
   C. bool price;
   D. char price;

PART B (60 MARKS)

QUESTION 1

This question tests algorithms and programming errors.

a) Define algorithm and pseudocode.

(2 marks)

b) Explain THREE types of programming errors and give one example of each.

(3 marks)

QUESTION 2

A program reads a student's score from 0 to 100 and displays the grade shown in Table 1.

Table 1. Grade Scale

Score	Grade
80-100	A
60-79	B
50-59	C
40-49	D
0-39	F

a) Draw a flowchart for the grade classification.

(5 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

(8 marks)

QUESTION 3

This question tests declarations, constants and expressions.

a) Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

(6 marks)

b) Write C++ statements to read age and price, calculate price after discount, and display the result.

(5 marks)

QUESTION 4

Study the following nested loop:

```
for (int i = 1; i <= 3; i++) {
    for (int j = 1; j <= i; j++) {
        cout << j;
    }
    cout << endl;
}
```

a) Trace the program and show the exact output.

(7 marks)

b) Identify TWO logic errors in a version that prints four rows but reuses j without resetting it. Rewrite the corrected code.

(8 marks)

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QUESTION 5

A shop stores the item prices shown in Table 1.

a) Declare an array named prices using the values in Table 1. Display all values.

Table 2. Item Prices

Item	Price (RM)
1	20.50
2	55.00
3	15.90
4	75.00
5	10.00

(6 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

(10 marks)

PART C (30 MARKS)

QUESTION 1

A catering company charges the meal prices shown in Table 1.

Table 3. Meal Price

Quantity Range	Price per Meal (RM)
50 - 99	11.50
100 - 199	10.50
200 and above	9.50

Write a complete C++ program that reads the customer name and quantity, validates the quantity, calculates the total, displays a receipt, and repeats until the user chooses to stop. At the end, display the total quantity and total sales.

(15 marks)

QUESTION 2

A cafe sells the items shown in Table 1. A 10% discount is given when the subtotal is at least RM50.

Table 4. Cafe Menu

Item	Price (RM)
Coffee	8.50
Tea	6.00
Juice	9.50

Write a complete C++ program that reads the item code and quantity, calculates the subtotal and discount, and displays a formatted receipt with the item name, quantity, subtotal, discount and amount to pay.

(15 marks)

END OF QUESTION PAPER

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